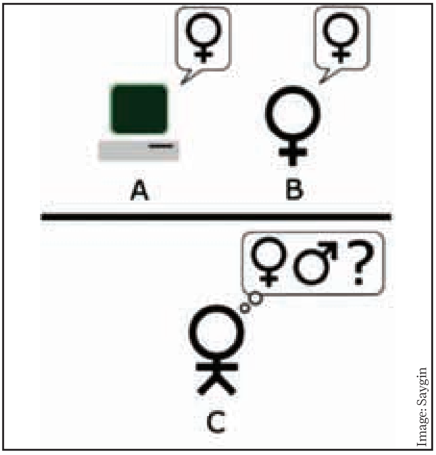
How is right and wrong determined in cases of logic? The word "think" throws up too many problems to be able to be used consistently for scientific determination, which we would all agree is necessary when trying to judge whether an artificial intelligence system is working or not. So how does one form a base test to adjudicate whether a system can be said to be an intelligent agent?

British computer scientist Alan Turing decided to redefine the parameter of what made a system an intelligent agent. Till 1950, the idea of artificial intelligence was to ask the question, "Can the machine think?" Turing came up with a new question in a 1950 paper titled Computing Machinery and Intelligence: "Are there imaginable digital computers which would do well in the imitation game?" The "imitation game" here is the ability of a machine to successfully imitate human thought and response, thus providing a much more measurable test for artificial intelligence systems.

His reasoning was that the efficacy of an artificial intelligence system lies in being able to be as human-like in its process of thinking as possible – as long as that end goal was being met, the mechanics of how it was achieved would be meaningless.

The two types of Turing Test

The "imitation game" that Turing referred to in his paper is an old party game, usually played by three players. For the sake of example, let's call them A, B and Interrogator. In the game, A is a woman, B is a man (or vice versa) and the Interrogator – who can't see either of them – has to guess the sex of both A and B by asking them a series of questions. It's all done through a cold text interface like a computer screen so as not to provide any hints.



The "standard interpretation" of the Turing Test, in which player C, the interrogator, is tasked with trying to determine which player - A or B - is a computer and which is a human. The interrogator is limited to using the responses to written questions in order to make the determination.